

Synpipe: A Containerized GUI Pipeline for Automated Synapse Detection and Analysis

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INTRODUCTION

Synaptic connections are fundamental to brain function, enabling communication between neurons and supporting learning and memory through synaptic plasticity (Graves et al., 2021). At the molecular level, AMPA receptor (AMPA) density and localization regulate synaptic strength.

Advances such as in vivo 2-photon imaging and fluorescently tagged AMPAR models (SEP-GluA2) have improved our ability to observe synaptic dynamics (Xu et al., 2023). However, analyzing the resulting large, noisy 3D datasets requires multiple tools — XTC for restoration and MR-CNN for segmentation — leading to fragmented, irreproducible workflows (Chen et al., 2025).

This fragmentation limits scalability and reproducibility, requiring manual integration across platforms with increasing variability across studies.

OBJECTIVE

- **Containerize each tool independently:** Package XTC and MR-CNN with all dependencies into isolated Docker containers to eliminate environment conflicts.
- **Unify via Hydra orchestration:** Connect containers into a single, configurable pipeline, enabling seamless data flow between processing steps.
- **Build a live GUI dashboard:** Develop a Flask-based web interface on the navon HPC cluster that lets researchers launch, monitor, and inspect pipeline runs in real time.
- **Validate output fidelity:** Confirm that containerized outputs are statistically identical to the original non-containerized workflow.

METHODOLOGY

Consolidating XTC restoration, MR-CNN segmentation, and post-processing into a single orchestrated framework

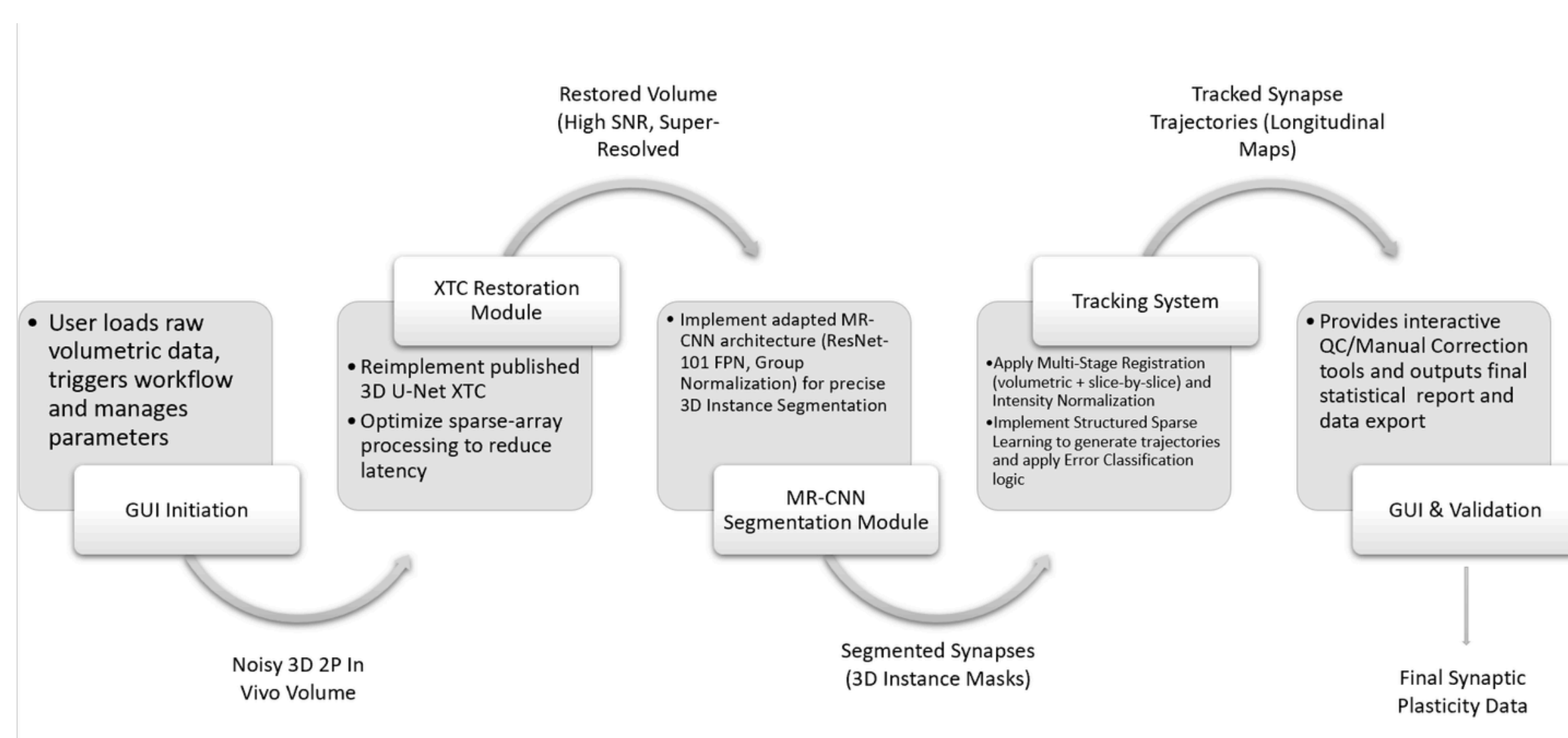


Figure 1. Integrated System Architecture of the Synpipe Platform. This diagram visualizes the unified workflow we aim to build to replace the current fragmented analysis methods. A user-friendly GUI will orchestrate the entire system, automating the transitions between the performance-optimized XTC Restoration Module and the adapted MR-CNN Segmentation Module to ensure high-throughput, reproducible analysis.

REFERENCES

- Chen Z, Coste GI, Li E, Huganir RL, Graves AR, Charles AS. Automatic detection of fluorescently labeled synapses in volumetric in vivo imaging data. bioRxiv [Preprint]. 2025 Jan 23;2025.01.22.634278. doi: 10.1101/2025.01.22.634278. PMID: 39896629; PMCID: PMC11785214.
- Xu, Y.K.T., Graves, A.R., Coste, G.I. et al. Cross-modality supervised image restoration enables nanoscale tracking of synaptic plasticity in living mice. Nat Methods 20, 935–944 (2023). <https://doi.org/10.1038/s41592-023-01871-6>
- Austin R Graves, Richard H Roth, Han L Tan, Qianwen Zhu, Alexei M Bygrave, Elena Lopez-Ortega, Ingie Hong, Alina C Spiegel, Richard C Johnson, Joshua T Vogelstein, Daniel J Tward, Michael I Miller, Richard L Huganir (2021) Visualizing synaptic plasticity in vivo by large-scale imaging of endogenous AMPA receptors eLife 10:e66809 <https://doi.org/10.7554/eLife.66809>
- Docker Inc. Docker Documentation. <https://docs.docker.com/>
- Meta AI. Hydra Documentation. <https://hydra.cc/>

ANALYSIS

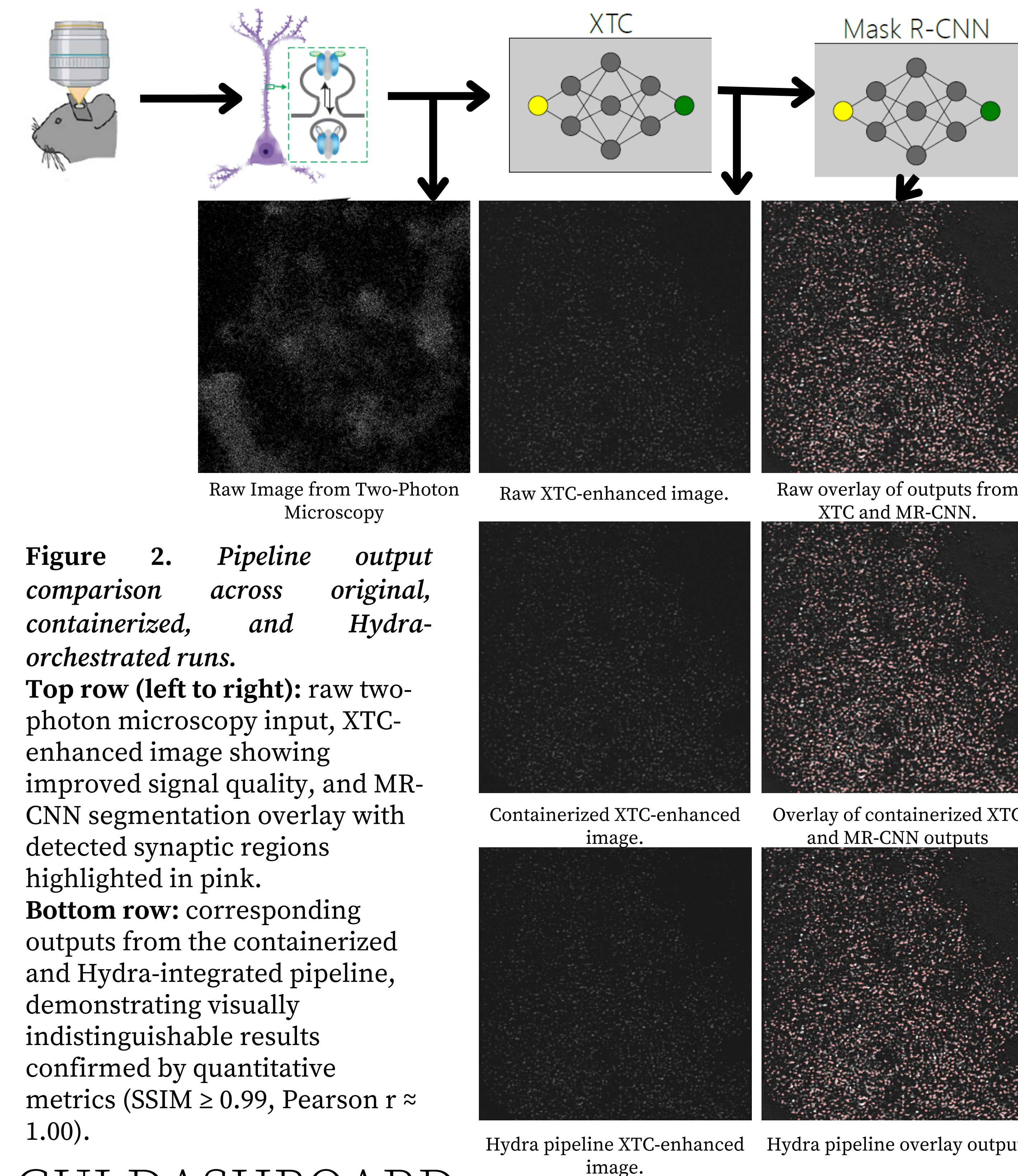


Figure 2. Pipeline output comparison across original, containerized, and Hydra-orchestrated runs.

Top row (left to right): raw two-photon microscopy input, XTC-enhanced image showing improved signal quality, and MR-CNN segmentation overlay with detected synaptic regions highlighted in pink.
Bottom row: corresponding outputs from the containerized and Hydra-integrated pipeline, demonstrating visually indistinguishable results confirmed by quantitative metrics (SSIM ≥ 0.99 , Pearson $r \approx 1.00$).

GUI DASHBOARD

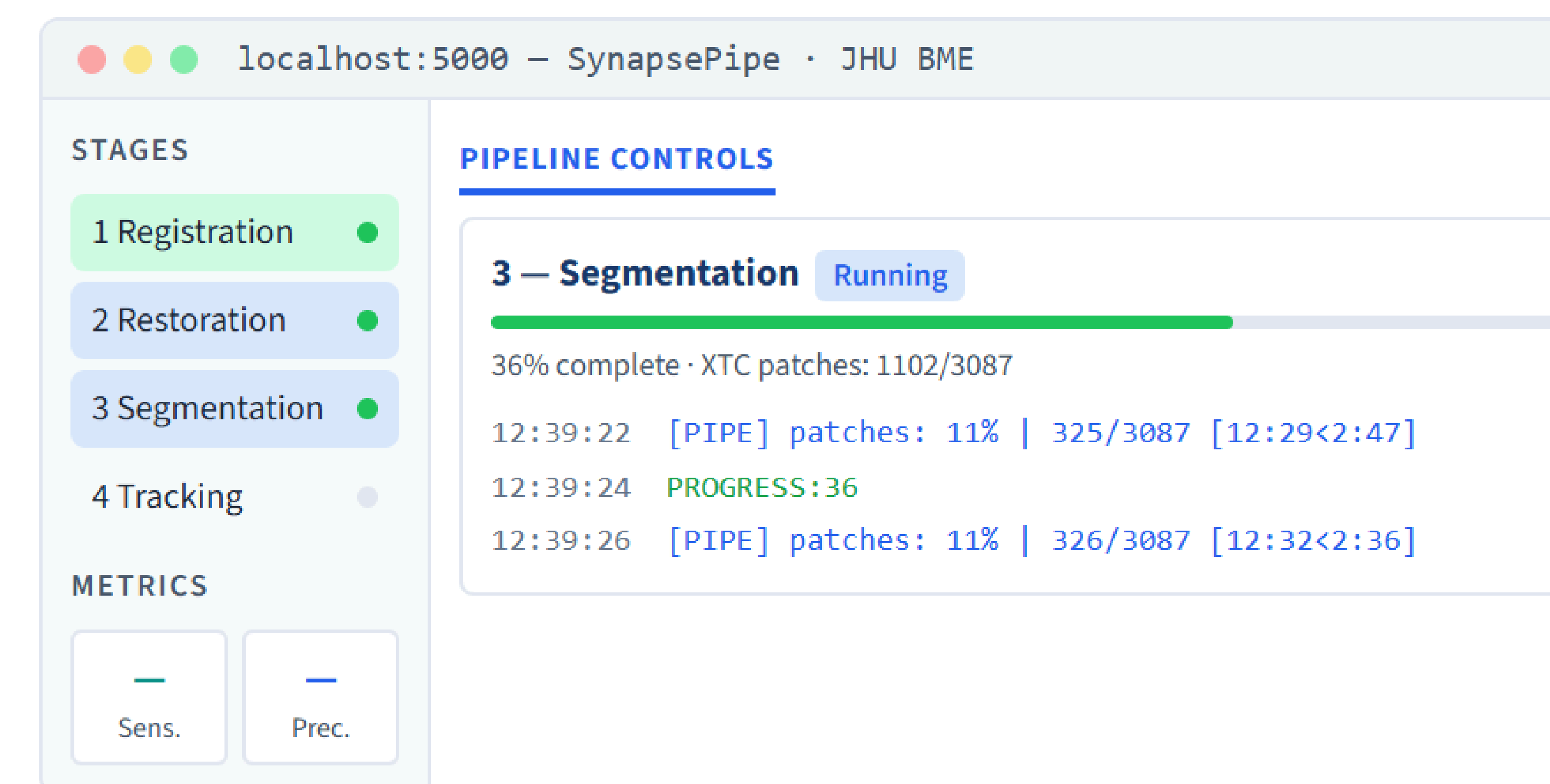


Figure 3. SynapsePipe GUI dashboard running live on navon HPC, accessible via browser at localhost:5000.

Table 1. Quantitative comparison of containerized and Hydra-orchestrated pipeline outputs against the original workflow.

Containerized Run Comparison				Hydra Comparison		
Metric	XTC Processed	Overlay	MRCNN Segmentatio	XTC Processed	Overlay	MRCNN Segmentation
Left size	1024x1024	1024x1024	1024x1024	1024x1024	1024x1024	1024x1024
Right size	1024x1024	1024x1024	1024x1024	1024x1024	1024x1024	1024x1024
Compared	1024x1024	1024x1024	1024x1024	1024x1024	1024x1024	1024x1024
Resized	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
SSIM	0.999955	0.993076	0.99054	0.999946	0.991638	0.989608
Pearson	1	0.997757	0.997413	0.99999	0.997189	0.997264
MSE	0	0.000188	0.000548	0	0.000232	0.000568
Shift row	0	0	0	0	0	0
Shift col	0	0	0	0	0	0
Shift mag	0	0	0	0	0	0
Shift err	1	1	1	1	1	1
Status	completed	completed	completed	completed	completed	completed

RESULTS & FINDINGS

- Pipeline successfully containerized and validated across all three output types
- SSIM ≥ 0.99 confirms containerization preserves output fidelity
- Pearson $r \approx 1.00$ with near-zero MSE and zero spatial shift across all comparisons
- Hydra pipeline matches original workflow with SSIM 0.9999 on XTC output
- GUI dashboard running live on navon A100 GPU — accessible via browser at localhost:5000
- Real-time log streaming enables researchers to monitor XTC patch processing without terminal access

CONCLUSION

By formulating an integrated pipeline, we aim to significantly reduce the complexity and manual effort required for synaptic imaging analysis. Through Hydra-based orchestration, users can specify input data and automatically execute sequential steps, such as image enhancement and segmentation, without manually intervening at each stage. This streamlined workflow improves efficiency, consistency, and usability compared to fragmented approaches.

In the future, we plan to extend this pipeline by incorporating a synaptic tracking program that operates after segmentation to analyze changes over time. Integrating this component will enable end-to-end synaptic analysis within a single framework, further enhancing the pipeline's capability to study synaptic dynamics and plasticity in a more comprehensive and automated manner.

FUTURE DIRECTIONS

- LAP-based synapse tracking across timepoints (in development)
- ITK-Elastix registration integration
- Docker Compose for one-command deployment